BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 287 · Run-length and dictionary coding

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. Define lossless compression.

Q2. What is a run in RLE?

Q3. What is a compression overhead?

Q4. For array [2,3,4,5], double values at indexes 1 and 3. State final array and sum.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

RLE — Encode repeated runs with counts.

run — Maximal consecutive equal symbols.

dictionary coding — Represent repeated strings by references.

dictionary — Mapping of codes to values/phrases.

overhead — Metadata/structure required by encoding.

New learning

Understand — Run-length encoding replaces each maximal run of the same symbol with a value and count; it works best for long consecutive repetitions.

Apply — Dictionary coding replaces repeated strings with dictionary references; the decoder needs the same dictionary, transmitted or reconstructed according to the scheme.

Evaluate — Overhead matters. Alternating characters can make naive RLE larger; long repeated phrases can be valuable for dictionary coding.

Worked example

Encode BBBBAAC as symbol/count pairs and verify decoding.

1. Predict first — Identify maximal consecutive runs: four B, two A and one C.

2. Model and check — Output each run as value/count and expand them back to verify exact reconstruction.

We Do · Practise with guidance

A dictionary maps index 1 to 'RED'. Encode 'RED RED' and decode the result.

```
Dictionary[1] = 'RED'
Encoded = (1,' ',1)
Decoded = 'RED RED'
```

Predict first

Use dictionary reference 1 for each repeated word while preserving the space separator.

Model and check

Decoder reads both references with its matching dictionary and reconstructs the original phrase.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

RLE-encode AAAAABBCCCDDDD with counts. [4]

Question 2

With dictionary 0='HELLO',1='WORLD', encode 'HELLO WORLD HELLO' using references and separators. [6]

Question 3

Explain why RLE can increase size for alternating data and why dictionary coding may help repeated long substrings. [6]

Exit Ticket · Answer before checking

RLE always shortens any input file.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Original can be reconstructed exactly.

Q2. Consecutive repeated instances of the same symbol.

Q3. Extra metadata needed to represent/decode data.

Q4. [2,6,4,10], sum 22; indexes are positions 1 and 3.

You Do pages · Independent questions

Q1. (A,5) [1], (B,2) [1], (C,3) [1], (D,4) [1].

Q2. 0 [1], space [1], 1 [1], space [1], 0 [1]; decoder with same dictionary recovers exact phrase [1].

Q3. No long runs [1], counts add overhead [2]; repeated substrings dictionary references [2], dictionary itself costs space [1].

Exit Ticket · Model answer

False. It may expand files with few repeated runs.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: